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B.Tech DEGREE EXAMINATION, NOVEMBER 2024

Fifth Semester

18CSC362J - COMPILER DESIGN

(For the candidates admitted from the academic year 2018-2019 to 2021-2022)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 100

PART - A (20 x 1 = 20 Marks) Answer ALL Questions

Marks BL CO PO

1. Match the following:

1 2 1 3

- | | |
|---------------------------|---------------------------|
| a. Push down automata | i) Regular expression |
| b. Finite State Machines | ii) Context free grammar |
| c. Syntax analysis | iii) Lexeme |
| d. Forward pointer | iv) Parse tree |
| A) a-ii; b-i; c-iv; d-iii | B) a-i; b-ii; c-iv; d-iii |
| C) a-ii; b-i; c-iii; d-iv | D) a-ii; b-iii; c-iv; d-i |

2. _____ is the maximum number of states that an ϵ -NFA can have on ϵ moves.

1 2 1 3

- | | |
|-------------|------|
| A) n | B) 0 |
| C) Infinite | D) 1 |

3. What is the length of the improper substring of "equivalent"?

1 2 1 3

- | | |
|-------|-------|
| A) 10 | B) 9 |
| C) 8 | D) 11 |

4. The _____ is realized by _____ operation.

1 1 1 1

- | | |
|--------------------------------|--------------------------------------|
| A) sequential execution, union | B) iteration, closure |
| C) branching, concatenation | D) conditional branch, concatenation |

5. Grammar that produces more than one parse tree for same sentence is _____.

1 1 2 2

- | | |
|--------------------|------------------|
| A) Ambiguous | B) Unambiguous |
| C) Complementation | D) Concatenation |

6. Which of the following statement is false?

1 1 2 2

- | | |
|--|--|
| A) Grammars are more powerful notation than regular expression | B) Regular expression can be described by a grammar |
| C) All Grammar Production are left recursive | D) Every regular language is a context free language, but not vice versa |

7. The grammar $S \rightarrow Sa \mid b$ is not suitable for predictive-parsing because the grammar has _____.

1 1 2 2

- | | |
|--------------------|------------------------|
| A) Left factoring | B) Left recursive |
| C) Right recursive | D) An operator grammar |

8. Which of the following derivations does a top-down parser use while parsing an input string? 1 1 2 2
 A) Leftmost derivation B) Leftmost derivation in reverse
 C) Rightmost derivation D) Rightmost derivation in reverse
9. The given grammar is for SLR 1 2 3 2
 $S \rightarrow (B)$
 $S \rightarrow \epsilon$
 $B \rightarrow AS$
 $A \rightarrow S$
 What is I0?
 A) $S' \rightarrow .S$ B) $S' \rightarrow .S$
 $S \rightarrow .(B)$ $S \rightarrow .(B)$
 $S \rightarrow .\epsilon$ $S \rightarrow .a$
 $B \rightarrow .AS$
 $A \rightarrow .S$
 C) $S' \rightarrow .S$ D) $S' \rightarrow .S$
 $S \rightarrow .(B)$ $S \rightarrow .(B)$
 $S \rightarrow .\epsilon$ $S \rightarrow .$
 $B \rightarrow .AS$
 $A \rightarrow .S$
10. Which grammar rules violate the requirements of an operator grammar? 1 1 3 2
 A) $S \rightarrow Ab$ B) $A \rightarrow A+D$
 C) $D \rightarrow cF$ D) $F \rightarrow BD$
11. If there is no handle on the stack then we have to use----- 1 1 3 2
 A) Reduce action B) Shift action
 C) Conflict removing strategy D) Shift- Reduce action
12. How many operations are required for translating the expression given below into quadruple representation? 1 3 3 3
 $(i*j)+(e+f)*(a*b+c)$
 A) 5 B) 6
 C) 7 D) 8
13. Which of the following is not a three-address code? 1 2 4 2
 A) $a = 5$ B) $b = a$
 C) $c = a + b$ D) $d = a + b - c$
14. Consider the following code: 1 2 4 2
 if (a
 { $b = 1 + 2$;
 $c = b + 2$;
 print(c);
 } Which of the following optimization can be applied?
 A) constant folding B) common sub expression elimination
 C) loop unrolling D) dead code elimination

15. The sequence of procedure calls of a program corresponds to which traversal of the activation tree? 1 1 4 1
 A) In order traversal B) Pre order traversal
 C) Post order traversal D) Level order traversal
16. The following code is an example of? 1 2 4 2

```
Void add_ten(int x)
{ return x + 10;
printf("value of x is %d", x);
}
```

 A) Redundant instruction elimination B) Unreachable code
 C) Flow of control optimization D) Reachable code
17. Which graph describes the basic block and successor relationship? 1 1 5 1
 A) Control Graph B) DAG
 C) Flow graph D) Hamiltonian graph
18. In algebraic expression simplification, $a = a + 1$ can simply be replaced by? 1 2 5 2
 A) a B) INC a
 C) DEC a D) MUL a
19. Ambiguous definitions for variables are more common during _____. 1 1 5 1
 A) Function calls B) Array declarations
 C) Register allocation D) Stack initialization
20. Which optimization techniques is used to reduce multiple jumps? 1 1 5 1
 A) Lateral optimization technique B) Peephole optimization technique
 C) Local optimization technique D) Code optimization technique

PART - B (5 x 4 = 20 Marks)

Answer ANY FIVE Questions

Marks BL CO PO

21. Distinguish the operational difference between compiler and interpreter. Also, list the tasks done by preprocessor. 4 2 1 2
22. Construct a DFA which can accept the strings starting and ending with same letter over the input $\Sigma = \{a, b\}$. Also, give its formal definition. 4 2 1 2
23. For the given grammar derive the string $(a+a)^*$ using Parse Tree and derivation. 4 3 2 3
 $S \rightarrow S+S \mid SS \mid (S) \mid S^* \mid a.$
24. Give the limitations of top down parsing. Also, eliminate left recursion from the following grammar. 4 2 2 3
 $Z \rightarrow Zde \mid u y w g \mid u h x$
25. Compute the Leading and Trailing for the given Grammar. 4 2 3 2
 $S \rightarrow S+S \mid S^*S \mid id$
26. Explain the translation scheme to produce three address code for assignment statements 4 2 4 2
27. Give the rules for determining the header in basic block. What are points in basic blocks? 4 2 5 2

PART - C (5 x 12 = 60 Marks)
Answer ALL the Questions

Marks BL CO PO

28 a. Convert the following RE= $a(a|b)^*abb$ to DFA using subset construction method and minimize it. 12 4 1 6

(OR)

b. Describe the various phases of compiler and trace it with the program segment (a: = b + real * 60). 12 4 1 6

29 a. Construct the Predictive parsing table for the given grammar and Parse the Input string (a,a) 12 4 2 2

S $\rightarrow$ (L) | a

L $\rightarrow$ L, S | S

(OR)

b. Check whether the grammar is LL (1) Or Not. 12 4 2 2

S $\rightarrow$ iEtS | iEtSeS | a

E $\rightarrow$ b

30 a. Construct LR(1) items and Parsing Table for the grammar. 12 4 3 2

S $\rightarrow$ CC

C $\rightarrow$ cC | d

(OR)

b. Consider the following Grammar 12 4 3 2

S $\rightarrow$ L=R

S $\rightarrow$ R

L $\rightarrow$ *R

L $\rightarrow$ id

R $\rightarrow$ L

Construct a SLR parsing table.

31 a. Explain the Back patching for the expression 12 3 4 3

$x < 100 \parallel y > 200 \&\& x \neq y$

(OR)

b. Write quadruples, triples and indirect triples for the expression: 12 3 4 3

$-(a*b)+(c+d)-(a+b+c+d)$

32 a. Discuss in detail about storage allocation strategies 12 3 5 3

(OR)

b. Generate the three-address code statement and construct the DAG representation for the expression $I = a + a*(b-c) + (b-c) * d$. 12 3 5 3
